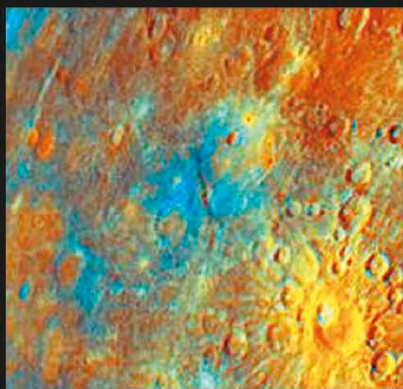
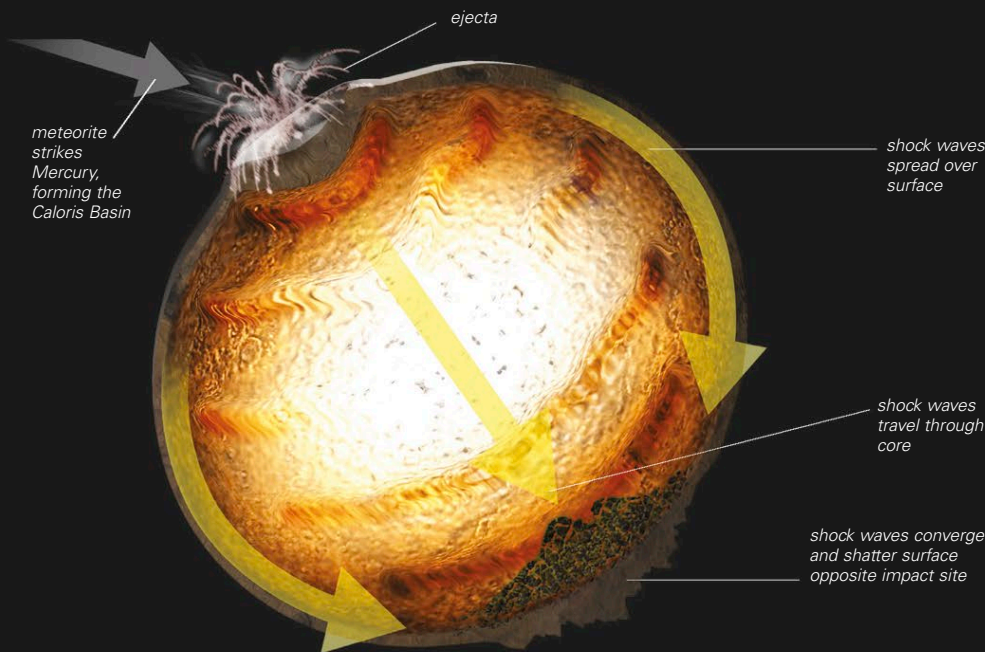


SURFACE FEATURES

Impact craters cover Mercury's visible surface. As the surface gravity is about twice that of the Moon, the ejecta blankets are closer to the parent craters and thicker than those found on the Moon. Large meteorite impacts have produced multiring basins. A particularly impressive example is the Caloris Basin. On the opposite side of the planet to the basin is a region of strange terrain produced by earthquakes resulting from the impact (see right). The craters are interspersed by at least two generations of flat plains of solidified basaltic lava, like the lunar maria. Fluid lava oozed gently out of vents in the crust and pooled in depressions. Eventually, most of the vents were covered by the lava. The MESSENGER space probe has photographed volcanic vents around the perimeter of the Caloris Basin, which are evidently the source of these lavas. Mercury's surface also has several ridges, which are up to 0.6–1.9 miles (1–3 km) high and 310 miles (500 km) long.

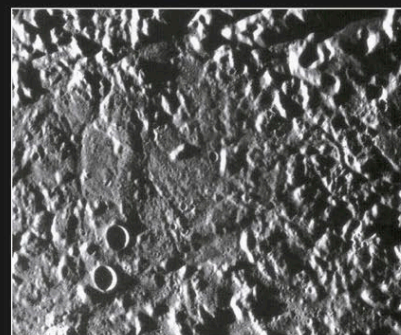


SURFACE COMPOSITION

In this false-color mosaic, yellow represents areas of the silicate crust that have been exposed by cratering, while the blue regions are younger volcanic rocks.

IMPACT SHOCK WAVES

A few minutes after the formation of the Caloris Basin, the shock waves generated by the impact came to a focus on the opposite side of the planet. This caused a massive upheaval over an area of 96,500 square miles (250,000 square kilometers), raising ridges up to 1.1 miles (1.8 km) high and 3–6 miles (5–10 km) across. Crater rims were broken into small hills and depressions.



CHAOTIC TERRAIN OPPOSITE THE CALORIS BASIN

